

Optoelectronics is Where the Expertise Lies!

Optoelectronics engineering involves roles that demand expertise and work in niche domains. That is why if you specialise and opt for a career in optoelectronics, you can actually enjoy favour in certain fields



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Optoelectronics engineering is a growing domain in India. Currently, it lacks mass opportunities, but is increasingly becoming lucrative for aspirants. The talent from optoelectronics stream is demanded across many sectors in the world of electronics—be it telecom, where optical fibre plays a major role, or VLSI.

Opportunities

Explaining the growth in demand for optoelectronics engineers, Alok Sinha, an industry professional with 22 years of experience, says, “The broadband revolution and the telecom boom in the country are surely the opportunities for optoelectronics engineers. The only reason why telecom service providers do not hire them is because the engineering

level they require is not at the level of optoelectronics engineers. The companies that actually need them are those which are responsible for ensuring fibre-optics networks and are last-mile providers, like a telecom company that has a large backhaul or a large telecom vendor like Huawei and ZTE that provides solutions in this space. Apart from them, companies like Tulip Network and even service providers like Sify and Hathway employ optoelectronics engineers.”

Unlike IT and telecom engineering, optoelectronics engineering is a pretty niche field. Optoelectronics engineers will find more opportunities in companies engaged in R&D than the field business. VLSI manufacturers would definitely need optoelectronics engineers on board to be able to find out new solutions and new environment infrastructure.

Unfortunately, in India, there are only a few manufacturers in fibre optics—a field where optoelectronics engineers are needed the most.

Raghu Panicker, country sales director, Mentor Graphics, says, “VLSI industry needs some of the finest optoelectronics engineering talent for making their products perfectly. We need people with background in optics and electronics. We need people with expertise in optoelectronics because when we talk of electronics design, we talk about semiconductors and circuit designs, where basically we create a layout and finally manufacture it. An electronics designer creates a layout and sends it for fabrication. His job is done when the data is taken to the magnetic tape and finally transferred to the foundry. What foundries do is create a mask for protection. They create mask for every layer. While creating a mask, engineers need a lot of knowledge about electronics in terms of how the masks are created. At the same time, they should also have an understanding of optics.”

Talent availability

Typically, it is the electronics and telecom engineers who have been filling in the gaps for optoelectronics engineers. But times are changing and so are the demands from various industries. Companies are now not ready to compromise. They prefer expert optoelectronics engineers only.

Sinha throws light on the availability of optoelectronics engineers in India: “The scenario has been gradually changing for the last five to six years. The fibre electronics, or the fibre-optic communication stream, has undergone